Nipun Dharmarathne

Department of Computer Engineering, University of Peradeniya.
+94 767593646 nipunsandeepakck@gmail.com linkedin.com/in/nipun-dharmarathne
github.com/NipunDharmarathne

I am a third year computer engineering undergraduate, interested in Embedded Systems, IOT, Computer Architecture and in all phases of Software Development. I am currently seeking for internship opportunities with determination to expand my horizons and to make a lasting impact on my future career.

Education

University of Peradeniya **November 2019 – Present**
Undergraduate in B.Sc. Engineering (Hons.) Computer Engineering *GPA 3.55/4.00*

Kingswood College, Kandy **2010 – 2018**
G.C.E. Advanced Level Examination *Z-Score: 1.8781*

Projects

Water Quality Monitoring System | Group | 2023
- A single device which integrates the hardware and software components needed to monitor water quality
- Technology: Mosquitto MQTT, Node-Red, MySQL, PlatformIO, Fritzing
- Techniques: Embedded systems, MQTT protocol

Automatically Categorizing Sinhala News Items from Selected News Sources | Group | 2023
- Automated classification of Sinhala news from chosen sources using ML and NLP techniques
- Technology: NLTK, Pandas, NumPy, Scikit-learn
- Techniques: Machine learning, NLP techniques

RISC-V 32-Bits Pipeline Processor with Cache and Memory | Group | 2023
- Designed and tested a RV32IM 5 stage pipeline processor with Verilog-HDL. Designed an assembler to compile machine language code to RISC-V Architecture
- Technology: Verilog-HDL
- Techniques: Computer Architecture, RISC-V

Smart Compost Management System | Group | October 2022
- A single device which integrates the hardware and software components needed to monitor and manage compost mixtures
- Technology: Heroku, GitHub, Figma, MongoDB, JSON, Arduino, Fritzing, EasyEDA
- Techniques: Embedded systems, Web, Computer and network security

Department Course Page for Undergraduate Students | Group | August 2022
- Developed undergraduate course page of the department website with more details, teacher-student interactions, and statistics of the previous years
- Technology: HTML, CSS, JavaScript, JSON, Jekyll, Bootstrap, Python, GitHub, Figma, Swagger
- Techniques: REST API, Version control

Student Results Management System | Group | May 2022
- Developed a mobile application to manage and analyze undergraduate students' marks
- Technology: Java, Android Studio
- Techniques: MVC architecture, Version control

8-Bits Single Cycle Processor with Cache and Memory | Group | March 2022
- An 8-bit single cycle CPU with associated memory hierarchy. The processor includes an ALU, register files, control logic, forwarding unit, data memory, data cache, instruction memory and instruction cache
- Simulated processor behaviour using Icarus Verilog and input and output signals were observed using GTKWave
- Technology: Verilog-HDL
- Techniques: RISC-V, Caching

Database System for University Gymnasium | Group March 2022
- A fully functional database to organize reservations, cancellations, borrowings and misplacement of equipment at a gym
- Technology: MySQL, PHP

Digital Clock | Individual August 2021
- A digital clock that prints the current time and date on the linux terminal. If a color is given as an argument to the program, the program will print a colored output
- Technology: C

Competitions and Achievements

ACES Hackathon **2023**
- An inter-university hackathon organized by ACES, University of Peradeniya. Participated under 'Other' category

ACES Coders v9.0 **2022**
- 12-hour algorithmic programming competition open to all undergraduates island wide
- Rank -29 (Out of 88 teams)

ACES Pre-Coders v9.0 **2022**
- 6-hour algorithmic programming competition open to all undergraduates of University of Peradeniya
- Rank - 5 (Out of 70 teams)

ACES Hackfest **2022**
- An inter-university hackathon organized by ACES, University of Peradeniya. Participated under 'Education' category

Inter School IT Competition **2015**
- Inter School IT Competition organized by ACES, University of Peradeniya
- Second Runners Up

International Mathematics and Science Olympiad **2011**
- Participated in the 8th International Mathematics and Science Olympiad (IMSO), held in Naga City, Philippines

National Mathematics and Science Olympiad
- Bronze medal - 2010, 2011 | Merit pass - 2012

Sri Lankan Mathematical Olympiad
- High Distinction - 2014, 2015, 2017 | Distinction - 2016

Experience

Voluntary
- Project Nanathambara: Contributed to deliver Python Beginners' Content to school students which was organized by ACES, University of Peradeniya
- Project Prathibana: Contributed to make papers and deliver the content to students in the sessions organized by Rotaract Club of University of Peradeniya

Rotaract Club
- Contributed in post designing, magazine designing, photography, volunteering and organising events

ACES and Hackers' Club
- Contributed in post designing and organising events

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL, Verilog HDL, ARM Assembly Language
Developer Tools: VS Code, Eclipse, Android Studio
Libraries: NumPy, Matplotlib, Pandas
Graphic designing: Adobe Photoshop, Adobe InDesign

Extra Curricular Activities

Member of the Rotaract Club of University of Peradeniya **2019 - Present**
- Digital media team member
- Organizing committee member of Project Captures, Prathibana and Paura

Member of the Design Team of ACES and Hackers' Club **2022 - Present**

Member of the Organizing Committee of Battlefield **2020**
- Badminton tournament organized by University of Peradeniya

President and Quiz Team Captain of Kingswood Science Society **2017 - 2018**

Member of the Chess Team of Kingswood College, Kandy **2010 - 2015**

Participated in the Road Race of Inter House Sports Meet of Kingswood College, Kandy **2013, 2014**

Certifications

Introduction to Cybersecurity Foundations - Coursera | **2023**

Operating System Foundations - Coursera | **2023**

Introduction to Web Development - Coursera | **2020**

Python Data Structures - Coursera | **2020**

References

Prof. Roshan G. Ragel | roshanr@eng.pdn.ac.lk

Head of Department, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka

Dr. Isuru Nawinne | isurunawinne@eng.pdn.ac.lk

Senior Lecturer, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka